Calculus II Lesson 19

Ratio Test for Infinite Series $\sum_{n=1}^{\infty} a_n$

a) If $\lim_{x\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

b) If $\lim_{x\to\infty}\left|\frac{a_{n+1}}{a_n}\right| > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

c) If $\lim_{x\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 1$, use another test.

1) $\sum_{n=1}^{\infty} \frac{1}{6^n}$ is convergent or divergent?

Let: $a_n = \frac{1}{6^n}$ and $a_{n+1} = \frac{1}{6^{n+1}}$

a) $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{\frac{1}{6^{n+1}}}{\frac{1}{6^n}} \right| = \lim_{n\to\infty} \left| \frac{6^n}{6^{n+1}} \right| = \underline{\ \ ?\ \ }$ Hint: $6^{n+1} = 6^n \cdot 6^1$

b) What does the Ratio Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \frac{1}{6^n}$?

2) $\sum_{n=1}^{\infty} \frac{n!}{3^n}$ is convergent or divergent?

Let: $a_n = \frac{n!}{3^n}$ and $a_{n+1} = \frac{(n+1)!}{3^{n+1}}$

a) $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{\frac{(n+1)!}{3^{n+1}}}{\frac{n!}{3^n}} \right| = \underline{\quad ? \quad}$

Hint: $3^{n+1} = 3^n \cdot 3^1$; $(n+1)! = (n+1)(n)(n-1)\cdots(3)(2)(1)$; $n! = (n)(n-1)\cdots(3)(2)(1)$

b) What does the Ratio Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \frac{n!}{3^n}$?

3) $\sum_{n=1}^{\infty} n\left(\frac{6}{5}\right)^{n}$ is convergent or divergent?

Note: $a_n = n\left(\frac{6}{5}\right)^{n}$ and $a_{n+1} = (n+1)\left(\frac{6}{5}\right)^{n+1}$

a) $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(n+1)\left(\frac{6}{5}\right)^{n+1}}{n\left(\frac{6}{5}\right)^{n}}\right| = \lim_{n\to\infty}\left|\frac{(n+1)}{n}\right| \cdot \lim_{n\to\infty}\left|\frac{\left(\frac{6}{5}\right)^{n+1}}{\left(\frac{6}{5}\right)^{n}}\right| = \underline{\quad ?\quad}$

Hint : $\left(\frac{6}{5}\right)^{n+1} = \left(\frac{6}{5}\right)^{n} \cdot \left(\frac{6}{5}\right)^{1}$

b) What does the Ratio Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} n\left(\frac{6}{5}\right)^{n}$?

4) $\sum_{n=1}^{\infty} \frac{n}{4^n}$ is convergent or divergent?

Let: $a_n = \frac{n}{4^n}$ and $a_{n+1} = \frac{n+1}{4^{n+1}}$

a) $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{n+1}{4^{n+1}}}{\frac{n}{4^n}}\right| = \lim_{n\to\infty}\left|\frac{n+1}{n}\right| \cdot \lim_{n\to\infty}\left|\frac{4^n}{4^{n+1}}\right| =$ ___?___ Hint: $4^{n+1} = 4^n \cdot 4^1$

b) What does the Ratio Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \frac{n}{4^n}$?

5) $\sum_{n=1}^{\infty} \frac{n^3}{3^n}$ is convergent or divergent?

Let: $a_n = \frac{n^3}{3^n}$ and $a_{n+1} = \frac{(n+1)^3}{3^{n+1}}$

a) $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{(n+1)^3}{3^{n+1}}}{\frac{n^3}{3^n}}\right| = \lim_{n\to\infty}\left|\frac{(n+1)^3}{n^3}\right| \cdot \lim_{n\to\infty}\left|\frac{3^n}{3^{n+1}}\right| = \underline{\quad ? \quad}$ Hint: $3^{n+1} = 3^n \cdot 3^1$

b) What does the Ratio Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \frac{n^3}{3^n}$?

6) $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n 2^n}{n!}$ is convergent or divergent?

Let: $a_n = \dfrac{(-1)^n 2^n}{n!}$ and $a_{n+1} = \dfrac{(-1)^{n+1} 2^{n+1}}{(n+1)!}$

a) $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\dfrac{(-1)^{n+1}2^{n+1}}{(n+1)!}}{\dfrac{(-1)^n 2^n}{n!}}\right| = \lim_{n\to\infty}\left|\frac{n!}{(n+1)!}\right|\cdot\lim_{n\to\infty}\left|\frac{2^{n+1}}{2^n}\right| = \underline{\quad ?\quad}$$

Hint: $2^{n+1} = 2^n \cdot 2^1$; $(n+1)! = (n+1)(n)(n-1)\cdots(3)(2)(1)$; $n! = (n)(n-1)\cdots(3)(2)(1)$

b) What does the Ratio Test say about the convergent/divergent of the series $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n 2^n}{n!}$?

7) $\displaystyle\sum_{n=1}^{\infty}\frac{n!}{n\cdot 3^n}$ is convergent or divergent?

Let: $a_n=\dfrac{n!}{n\cdot 3^n}$ and $a_{n+1}=\dfrac{(n+1)!}{(n+1)\cdot 3^{n+1}}$

a) $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{\dfrac{(n+1)!}{(n+1)\cdot 3^{n+1}}}{\dfrac{n!}{n\cdot 3^n}}\right|=\lim_{n\to\infty}\left|\frac{(n+1)!}{n!}\right|\cdot\lim_{n\to\infty}\left|\frac{n}{n+1}\right|\cdot\lim_{n\to\infty}\left|\frac{3^n}{3^{n+1}}\right|=\underline{\quad ?\quad}$$

Hint: $3^{n+1}=3^n\cdot 3^1$; $\quad(n+1)!=(n+1)(n)(n-1)\cdots(3)(2)(1)$; $\quad n!=(n)(n-1)\cdots(3)(2)(1)$

b) What does the Ratio Test say about the convergent/divergent of the series $\displaystyle\sum_{n=1}^{\infty}\frac{n!}{n\cdot 3^n}$?

8) $\sum_{n=0}^{\infty} \frac{3^n}{(n+1)^n}$ is convergent or divergent?

Let: $a_n = \frac{3^n}{(n+1)^n}$ and $a_{n+1} = \frac{3^{n+1}}{\left[(n+1)+1\right]^{n+1}} = \frac{3^{n+1}}{(n+2)^{n+1}}$

a) $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{\frac{3^{n+1}}{(n+2)^{n+1}}}{\frac{3^n}{(n+1)^n}}\right| = \lim_{n\to\infty} \left|\frac{3^{n+1}}{3^n}\right| \cdot \lim_{n\to\infty} \left|\frac{(n+1)^n}{(n+2)^{n+1}}\right| = \lim_{n\to\infty} \left|\frac{3^{n+1}}{3^n}\right| \cdot \lim_{n\to\infty} \left|\frac{(n+1)^n}{(n+2)^n} \cdot \frac{1}{(n+2)}\right| = \underline{\quad ? \quad}$

Hint: $3^{n+1} = 3^n \cdot 3^1$; $(n+2)^{n+1} = (n+2)^n \cdot (n+2)^1$; $\lim_{n\to\infty} \frac{(n+1)^n}{(n+2)^n} = \lim_{n\to\infty} \left(\frac{n+1}{n+2}\right)^n = \frac{1}{e}$

b) What does the Ratio Test say about the convergent/divergent of the series $\sum_{n=0}^{\infty} \frac{3^n}{(n+1)^n}$?

9) $\sum_{n=1}^{\infty} \frac{7^n}{2^n+1}$ is convergent or divergent?

Let: $a_n = \frac{7^n}{2^n+1}$ and $a_{n+1} = \frac{7^{n+1}}{2^{n+1}+1}$

a) $$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{\frac{7^{n+1}}{2^{n+1}+1}}{\frac{7^n}{2^n+1}}\right| = \lim_{n\to\infty}\left|\frac{7^{n+1}}{7^n}\right| \cdot \lim_{n\to\infty}\left|\frac{2^n+1}{2^{n+1}+1}\right| = \underline{\quad ? \quad}$$

Hint: $7^{n+1}=7^n \cdot 7^1$; $$\lim_{n\to\infty}\left(\frac{2^n+1}{2^{n+1}+1}\right) = \lim_{n\to\infty}\frac{\left(2^n+1\right)/2^n}{\left(2^{n+1}+1\right)/2^n} = \lim_{n\to\infty}\frac{(1+1/2^n)}{(2+1/2^n)} = \frac{1}{2}$$

b) What does the Ratio Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \frac{7^n}{2^n+1}$?